
Discussion: class design

Important concepts and structures

We'll use the following concepts to explain what is wrong with the design and what structure can be used to improve the design

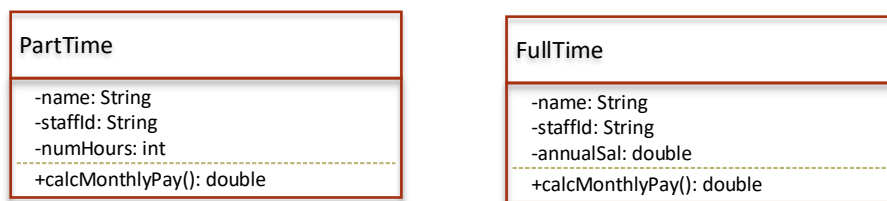
- Duplication
- Code reuse
- Tight coupling
- Cohesion
- Efficient design/code
- Classes, inheritance and interfaces

Question

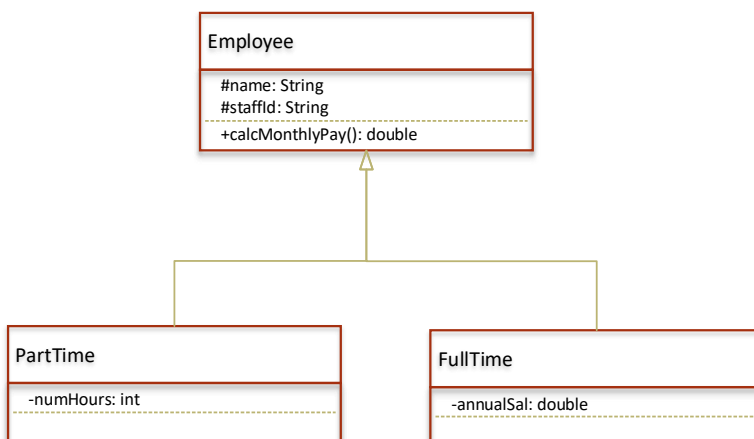
In each of the following, state:

- What is wrong with the design and how it can be improved?
- Which structure is appropriate?

1.1.



1.2 Note: Employee, PartTime and FullTime are concrete classes



1.3

Lecturer
-lid: String -name: String +insert(Lecturer):boolean +delete(Lecturer):boolean +getAllLecturers(): Lecturer[] +search(String):Lecturer +update(Lecturer):boolean

1.4

BankAccount
-holderName: String -holderId: String -holderAddress: String -accNo: String -bal: double +checkBalance(BankAccount ba): double +deposit(BankAccount ba, double amount): double +withdraw(BankAccount ba, double amount): double

Answers

1.1

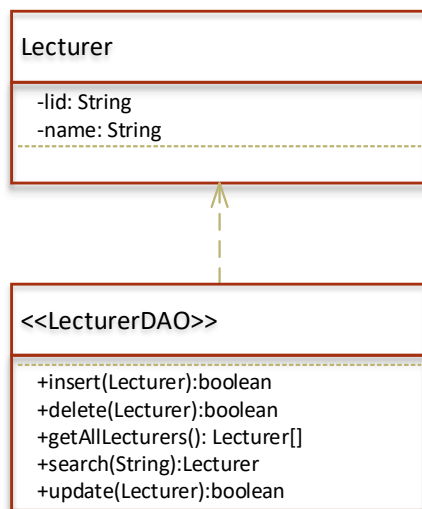
- a. What is wrong? Duplication
- b. How to improve: inheritance; subclasses inherit the methods and remove duplication

1.2

- a. What is wrong? Redundancy (calcMonthlyPay() will be implemented)
- b. How to improve: make calcMonthlyPay() abstract and Employee abstract; leave subclasses to override

1.3.

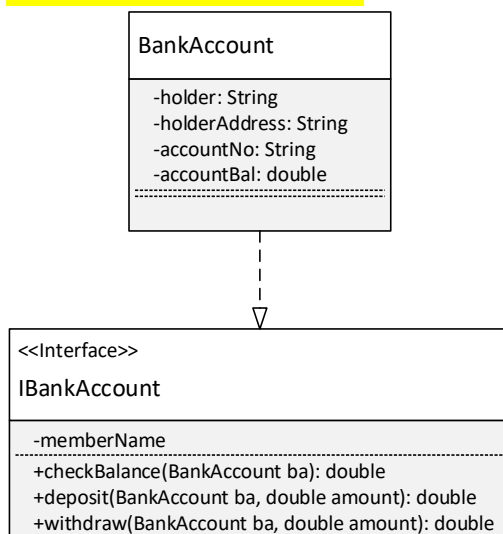
- a. What is wrong: Tight coupling between object and the methods which act on it
- b. How to improve: separate object and methods which act on it (create POCO for Lecturer object and an interface for the methods and show interaction).



1.4

- a. What is wrong: Tight coupling between object and the methods which act on it
- b. How to improve: separate object and the methods which act on it (create POCO for BankAccount and interface)

NOTE: this answer is incorrect



Correct

- a. What is wrong: Tight coupling in BankAccount (contain information about holder and bank account)
- b. How to improve: create POCO for Holder and BankAccount and interface and show interaction